



Smart Hotel

Web-Based Hotel Management System

Project Proposal

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1.Introduction.

The Hotel Management System (HMS) is a desktop-based software built for Windows computers. It will help to automate all the main tasks in a hotel like booking rooms, guests check ins and check outs, handling payments, managing food orders, and creating reports. These tasks are done with paper or simple spreadsheets right now. They can cause mistakes and delays especially when the hotel is busy.

At the beginning, the system will include around 100 room entries, 100 to 150 customer profiles, and about 30 to 40 food or service items. As the hotel grows, we expect it to develop 10,000 customer records and many daily transactions. So, we will make the database queries better from the start to keep the system fast, even with lots of data. The software will support different types of users like receptionists, restaurant staff, and managers, with each person only seeing what they need to do their job. There are some challenges like having only basic computers, users who are not fluent in technology knowledge and the need for the system to be easy, fast, and secure. The hotel serves both local and foreign guests and works with suppliers like food and cleaning services. Their information will also be stored in the system.

2.Motivation of the Project.

Hotel stakeholders face some challenges like delays, mistakes, and a lot of wasted resources when they manage hotel operations with manual or semi digital systems. It becomes more important to move toward a fully digital and efficient system that keeps operations smoothly when hotels expand and deal with more rooms and guests with higher expectations.

- Make smooth and speed up day today hotel operations.
- Make hotel staff more productive and happier in their roles and encourage to engage in their duties more efficiently.
- Eliminate paper-based processes and reducing environmental and operational waste like paper waste, ink waste etc.
- Automatically generate reports easily to gain clearer insights into the business and make smarter decisions.
- Improve guest experience through faster service and clearer communication.
- Enable easy tracking of bookings, payments, services, and inventory.
- Make easy to share data between departments without duplication.

3. Objectives of the project.

- **Automation**

The major activities in a hotel like room bookings, guest check-ins and check-outs, billing for food and handling daily admin work can all be done automatically. This saves time and makes everything easier for the staff.

- **Efficiency**

The system helps to avoid errors and delays when managing guest details, room availability, and staff schedules by decreasing manual work.

- **Accessibility**

The system is designed to use easily even for staff members who may not be having a fluent knowledge on technology. It makes more practical for everyday hotel operations.

- **Reporting**

It creates helpful reports and summaries on things like room occupancy, income, and food sales. This gives hotel managers a clear picture of how the hotel is doing.

- **Security**

Only authorized staff can access sensitive data, keeping hotel operations safe and secure with login controls and role-based permissions.

4.scope

4.1 In-scope (What is included in the system)

- Let guests to book and cancel rooms
- Handle check-in and check-out easily
- Show which rooms are free or reserved.
- Create bills and accept payments
- Collect guest feedback and reviews
- Let staff log in with different access levels
- Automatically create reports (like income and bookings)
- Keep track of hotel items and supplies
- Store records digitally (no need for paper)
- Send email or SMS updates to guests

4.2 Out-of-Scope (What is not included)

- Managing food orders or restaurant services.
- Connecting with travel agency websites
- Designing or building the hotel's public website
- Making a mobile app
- Full staff salary or HR system
- Controlling room devices like lights or air conditioning
- Using different languages in the system
- Linking with things like smart locks or security cameras

5. Specific Requirements

5.1 External Interfaces

The Hotel will use the standard input/output devices for a personal computer. This includes the following:

- Keyboard
- Mouse
- Monitor

5.1.1 User Interface

The user interface screens are described in the following table.

Home Page	Welcome user with an overview, featured rooms, search bar and login/ register buttons
Map View	Shows hotel location on a map with pins, help user locate hotel easily
Login/ Register	Allows user registration and login using email or third-party login
User Dashboard	Shows user profile, booking history, trip details

Hotel Details Page	View hotel description, facilities (AC, Wi-Fi, etc.), reviews, photo gallery
Room Types Page	Displays available room categories with prices, features, and images
Room Booking Page	User selects room, stay dates, guest count, and adds special requests
Payment Page	User selects payment method and enters card details
Booking Confirmation	Shows booking summary with invoice, refund policy, and confirmation email sent
Cancellation Page	Allows user to cancel booking and shows refund eligibility based on timing
Review & Ratings Page	Users can rate their stay and write feedback about the room and service
Discounts & Offers Page	Shows current deals, coupon codes, price comparisons
Admin Login Page	Login for hotel administrators
Admin Dashboard	Admin Dashboard
Room Management	Admin updates room info, price, and availability
User Management	Admin views and updates user accounts and booking history
Gallery Page	Hotel photos, room photos, and virtual tours
Settings Page	Allows language and currency preference selection

Contact / Help Page	User sends queries or complaints; links to FAQ
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5.1.2 Software Interfaces

- Email Service API
- Database API – Local database.

5.1.3 Hardware Interfaces

- Web Servers – Local web host
- User devices – Compatible with standard desktop computers and mobile devices.

5.1.4 Communication Interfaces

- HTTP/HTTPS for secure web communication.
- API for integration with external systems (Email services).

5.2 Functional Requirements

5.2.1 Login

1. User registration
2. User authentication
3. Forgot password
4. Password reset

5.2.2 User Roles and Permissions

1.Admin

- Full control over the system including user management.
- Payment gateways update

2.Users

- Register to website
- Picks the available room
- Can change profile details (Email, Phone number)

5.2.3 Booking Creation

1. Booking availability - Developer will provide room availability.

(Available or non-available)

2. Booking modifications - Can change rooms before the scheduled date

5.2.4 A Room Delete for the system temporally.

1. In the Past, another person booking that room.
2. when hotel staff determining that it is in an unsuitable condition for use. (Broken furniture, Damaged electrical items, Wall or ceiling damage etc.)

5.2.5 Room Booking

1. Room selection (Room category/No of rooms available)
2. Booking confirmation message via email (After payments)
3. View room booking history
4. Cancel Bookings

5.2.6 Payments

1. Payment method

Required Advance Payment Based on Booking Date

Time Before Check-in	Advance Payment Required
More than 30 days	20%
30-20 days	50%
Less than 20 days	100%

Refund Eligibility Based on Cancellation Time

Time Before Check-in	Refund on Cancellation
More than 30 days	80% Refundable
30-20 days	70% Refundable
20-7 days	60% Refundable
7-3 days	40% Refundable
Less than 3 days	No refund

2. Transaction Records
3. Receive confirmation of the payment. (Email)
4. Support online payments

5.2.7 Filtering rooms

1.Can filtering rooms as desired - Booking by ac/non-ac, single bed/double bed, Wi-Fi/non-Wi-Fi

5.2.8 Searching rooms

1.Checking available rooms

5.2.9 Sorting rooms

1.Based on price - can be selected according to the user's income.

2.Based on customer rating – can choose according to the highest customer rating.

5.3 Nonfunctional Requirements

5.3.1 Performance Requirements

1. Response Time - Time period of loading that page. The system should respond to quickly.

2.Error rate - system should maintain an error rate of less than.

3.Throughput - The system must be able to process continue under heavy load.

5.3.2 Logical Database Requirements

1. User Accounts

1. User ID
2. Full name
3. Email
4. Password
5. Contact number
6. Address
7. Full name

2. Room Details

- 1.number of rooms
2. facilities – Ac or non-Ac, etc.
- 3.Beds per room

3. Room Booking

1. Room ID
2. Booking details (Available or non-available)

4. Transaction Details

1. Transaction ID
2. Payment details
3. Payment status (Completed/Pending/Failed)

5.3.3 Reliability

1. User's reviews
2. Regular Backups

5.3.4 Availability

1. System Uptime - The platform must be available in 24 hours.
2. Hosting - The system should be deployed on a reliable hosting provider or cloud platform.

5.3.5 Security

1. User Authentication – opt verification with mobile numbers
2. ID verifications
3. Password protections
4. Encrypted Data
5. Secure Payment Process

5.3.6 Maintainability

1. Code Documentation - The codebase must be well-documented.
2. Version Control - GitHub must be used to manage code updates, and support collaboration.

5.3.7 Portability

1. Database Portability - Can application to work with different types of databases.
2. Browser Independence – The ability of software to run different web browser without any errors.
3. Cloud/Hosting Flexibility - The ability of software to run different hosting platforms without any errors.

6. Critical Functionalities.

- **Room Booking and Availability Management**

The room availability always should be updated to avoid double bookings as it leads to make customers uncomfortable.

- **Secure Login and Role-Based Access**

The system should be allowed only authorized users to access certain features considering their roles.

- **Payment and Billing System**

Handle all payments considering discounts and generate accurate bills for rooms and services.

- **Data Backup and Recovery**

Automatically save data and restore it when the system is crashed or data is accidentally deleted.

- **Report Generation and Summaries.**

Generate clear reports on bookings, income, and feedback for management.

7. What Hardware and software needed?

Hardware needed:

To run the system,

- ❖ **Server:** A dependable server to run the database and app. This can be a real machine or cloud service.
- ❖ **Client Computers:** Desktops or laptops to access the system for staff and managers
- ❖ **Networking Devices:** Router, switches, and stable internet connection for communication between clients and server.
- ❖ **Backup Storage:** External hard drives or cloud storage to back up data regularly to prevent data loss.
- ❖ **Barcode/QR Code Scanner (optional):** For quick check-ins or inventory management.
- ❖ **Printers:** For printing invoices, reports, or receipts.

To test the System,

- ❖ **Test Machines:** Additional PCs or laptops; or use virtual machines via VirtualBox or VMware.
- ❖ **Mobile Devices/Tablets:** If mobile testing is needed iPhone, Android phones or iPads are needed.
- ❖ **Network Simulation Tools:** Tools like Net Limiter or WANem to test how the system works with different internet speeds.

Software needed

To Develop & Run the System:

- **Programming Languages:**
 - **Frontend:** HTML, CSS, JavaScript (with frameworks like React, Angular, or Vue.js)
 - **Backend:** Python, Java (Spring), PHP (Laravel) or Node.js (JavaScript)
 - **Database Query Language:** MySQL
- **Web Development Software / Frameworks:**
 - **Frontend frameworks:** React.js, Angular, or Vue.js
 - **Backend frameworks:** Django, Flask, Spring Boot, Express.js, Laravel
 - **Code Editors / IDEs:** Visual Studio Code, IntelliJ IDEA, Eclipse, or PyCharm
- **Web Servers:**
 - Apache HTTP Server, Nginx, or Microsoft IIS
- **Version Control:** Git (with GitHub, GitLab, or Bitbucket)

8.Gnatt chart

[illegible]

9.Team Members.

Name	Role	Registration No	Index No
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